

GEN(NortonLifelock)

Gen Digital Inc. (formerly Symantec Corporation and NortonLifelock) is a multinational software company co-headquartered in Tempe, Arizona and Prague, Czech Republic. The company provides cybersecurity software and services. Gen is a Fortune 500 company and a member of the S&P 500 stock-market index. The company also has development centers in Pune, Chennai and Bangalore. Its portfolio includes Norton, Avast, LifeLock, Avira, AVG, ReputationDefender and CCleaner.

Hiring Role: Software Engineer Intern

The hiring process had 4 rounds.

- Online Test
- 2 Technical Rounds
- HR Round

Platform: DoSelect

Duration: 2 hours

Online Round: [30/09/2024]

It consisted of three coding question of medium level difficulty

Problem 1: Given a number n , convert it to its binary representation and store each binary digit in a linked list. The head of the linked list should contain the most significant bit (highest power of 2), and the head node cannot be 0.

The task is to return the head of the linked list containing the binary form of n .

This is a problem about converting a number to binary and storing the binary digits in a linked list format.

Sample Input: 10

Sample Output:

1
0
1
0

Problem 2: You are given a string s of length n . Find the length of longest substring whose sum is divisible by 3 and if no such substring exists then return 0.

Sample Input: 54641

Sample Output: 4

Problem 3: You are given a string *s* representing a number and an integer *k*. You can erase at most *k* digits from the number. Your task is to return the smallest possible number that can be obtained by removing up to *k* digits from *s*.

This problem is about minimizing a number by strategically removing up to *k* digits from the original string.

Sample Input: 4529350

3

Sample Output: 2350

A total of 83 people were shortlisted for interview

Technical Round 1: [21/10/2024]

In the first round of the interview, I was asked to introduce myself, followed by questions on Data Structures and Algorithms (DSA), Database Management Systems (DBMS), and Object-Oriented Programming (OOPS).

Coding Questions:

I was asked to write code for problems such as:

- Checking if a given array can be formed into a Binary Search Tree (BST)
- Reversing an array in place
- Buy and sell stock problem
- Maximum product subarray
- Pairwise swap of elements in a linked list

Data Structures and Concepts:

The interviewer asked basic questions about linked lists, trees, graphs like differences between trees and graphs and types of trees, etc.

Following this, I was asked 3-4 practical application questions.

Example:

A question, where a text message on WhatsApp can be edited and undone, what data structure can be used, I initially suggested using a stack. However, the interviewer pointed out that the data shouldn't be deleted as it can be redone. I then proposed using a linked list, where the edited message could be added as the next element, and when changes are undone, a pointer could move to the previous element, or to the next if redone. However, the interviewer asked to avoid pointers, so I came up with a solution using two stacks: one to store the current edits, and the other to store elements when they are undone, allowing for an undo-redo functionality without data deletion and she was satisfied with this

DBMS:

I was asked to write a query to find the second largest salary from a table.

OOPS:

The interviewer asked me to explain a concept in OOPS that made coding easier and how it differs from regular procedural code.

After the interview, 35-40 candidates were shortlisted for the next round.

Technical Round 2:

This was also eliminatory round.

In this round of the interview, I was asked questions on topics like Operating Systems (OS), Data Structures (DS), Object-Oriented Programming (OOPS), and projects mentioned in my resume.

Data Structures (DS):

The interviewer focused on testing my understanding of data structures, rather than definitions or explanations.

I was also asked about adjacency lists and some tree-related questions.

Object-Oriented Programming (OOPS):

He asked me to apply OOP concepts to a building with multiple lifts and explain the communication between classes. Initially, I was thinking about how to apply inheritance and the four pillars of OOP between classes. However, the interviewer was looking for a basic answer—just the class names, which turned out to be "class Lift" and "class Floor." 😊 (I was trying to overcomplicate it with code!). He laughed for overcomplicating.

Operating Systems (OS):

I was asked a question related to process scheduling.

I was selected for the next round.

HR Round:

It was an eliminatory round, the interviewer asked me various questions related to my resume, including the tech stacks I used in my projects, club activities, and ongoing work in the clubs. I was also asked how many LeetCode questions I had solved, when I started using LeetCode, and what I am currently learning. I mentioned that I am learning Java, as I am primarily a C++ programmer, and many companies expect Java for cloud development.

When asked about the next field I plan to learn (the interviewer preferred something other than cybersecurity since the company was Norton, which focuses on cybersecurity), I mentioned Artificial Intelligence (AI). I explained how AI can be both a cybersecurity threat (due to things like deepfakes) and a tool to protect digital users.

Technical Questions:

Difference between IPv4 and IPv6

What is cache?

Explain shared lock and exclusive lock in DBMS as if explaining to a kid.

In the middle of the interview, the interviewer introduced himself and asked which team I would like to work with, then explained the different teams in the company. Before concluding, the interviewer advised me to specialize in a specific field, as many candidates had similar tech skills on their resumes, making it difficult to stand out.

A total of 21 people were selected at the end.

Tips:

Be strong in DSA and core CS concepts like DBMS, OS, OOPS

Practice leetcode regularly

Be bold and confident while answering

Ask for hints if you aren't able to think of a solution rather than saying I don't know.

All The Best!!!

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